

Roozbeh Mottaghi

Email: roozbehm@gmail.com
Webpage: roozbehm.info
Google Scholar: <https://scholar.google.com/citations?user=CCV58dgAAAAJ>

RELEVANT PROJECTS

- [PARTNR](#) – A suite of models and benchmarks for evaluation of agentic planning and reasoning in human-agent collaboration.
- [ALFRED](#) – A benchmark for interpreting grounded instructions for everyday tasks.
- [OK-VQA](#) / [A-OKVQA](#) – A benchmark for visual question answering that evaluates world knowledge and commonsense reasoning.
- [InfLevel](#) – A benchmark for physical reasoning capabilities of AI models.
- [Navigation benchmarking](#) – Defining evaluation protocols and benchmarks for embodied navigation.
- [GOAT-Bench](#) – Evaluation of multi-modal lifelong navigation.
- [Habitat 3.0](#) / [AI2-THOR](#) / [ProcTHOR](#) – Frameworks developed to evaluate various planning, reasoning, 3D visual understanding, and embodied problems.

PROFESSIONAL EXPERIENCE

FAIR at Meta

Senior AI Research Scientist Manager

Aug 2022 to now

Leading a cross-org team of 20+ to develop and benchmark agentic planning and reasoning models for human-agent collaboration.

Paul G. Allen School of Computer Science & Engineering University of Washington

Affiliate Associate Professor

Sep 2019 to now

Co-advising PhD students on computer vision and embodied AI problems.

Allen Institute for Artificial Intelligence (AI2)

Research Manager

Jan 2020 to Aug 2022

Senior Research Scientist

May 2017 to Jan 2020

Research Scientist

Jan 2015 to May 2017

Research Manager of the Perceptual Reasoning and Interaction Research (PRIOR) team, leading 10+ researchers on modeling and benchmarking for Embodied AI and computer vision, including navigation, physical reasoning, instruction following, multimodal models, and knowledge-based QA.

Vicarious

Senior Researcher

Oct 2014 to Dec 2014

Computational Vision and Geometry Lab, Stanford University

Postdoctoral Researcher

Sep 2013 to Sep 2014

EDUCATION

University of California-Los Angeles, Los Angeles, California USA

PhD., Computer Science

Sep 2008 to Sep 2013

(Refer to Google Scholar for the full list)

M. Patel, X. Puig, R. Desai, **R. Mottaghi**, S. Chernova, J. Truong, A. Rai. **ADAPT: Actively Discovering and Adapting to Preferences for any Task**, Conference on Language Modeling (CoLM), 2025.

(*alphabetical order*) M. Chang, G. Chhablani, A. Clegg, M. Dallaire Cote, R. Desai, M. Hlavac, V. Karashchuk, J. Krantz, **R. Mottaghi**, P. Parashar, S. Patki, I. Prasad, X. Puig, A. Rai, R. Ramrakhya, D. Tran, J. Truong, J. M. Turner, E. Undersander, T.-Y. Yang, **PARTNR: A Benchmark for Planning and Reasoning in Embodied Multi-agent Tasks**, International Conference on Learning Representations (ICLR), 2025.

M. Khanna, R. Ramrakhya, G. Chhablani, S. Yenamandra, T. Gervet, M. Chang, Z. Kira, D. S. Chaplot, D. Batra, **R. Mottaghi**, **GOAT-Bench: A Benchmark for Multi-Modal Lifelong Navigation**, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

X. Puig*, E. Undersander*, A. Szot*, M. Dallaire Cote*, T.-Y. Yang*, R. Partsey*, R. Desai*, A. W. Clegg*, M. Hlavac, S. Y. Min, V. Vondruš, T. Gervet, V.-P. Berges, J. M. Turner, O. Maksymets, Z. Kira, M. Kalakrishnan, J. Malik, D. S. Chaplot, U. Jain, D. Batra, A. Rai[†], **R. Mottaghi**[†], **Habitat 3.0: A Co-Habitat for Humans, Avatars and Robots**, International Conference on Learning Representations (ICLR), 2024.
(*: Core team, [†]: Project leads)

S. Yenamandra, A. Ramachandran, K. Yadav, A. Wang, M. Khanna, T. Gervet, T.-Y. Yang, V. Jain, A. Clegg, J. Turner, Z. Kira, M. Savva, A. Chang, D. Chaplot, D. Batra, **R. Mottaghi**, Y. Bisk, C. Paxton, **HomeRobot: Open-Vocabulary Mobile Manipulation**, Conference on Robot Learning (CoRL), 2023.

M. Deitke, E. Vanderbilt, A. Herrasti, L. Weihs, J. Salvador, K. Ehsani, W. Han, E. Kolve, A. Farhadi, A. Kembhavi, **R. Mottaghi**. **ProcTHOR: Large-Scale Embodied AI Using Procedural Generation**, Advances in Neural Information Processing Systems (NeurIPS), 2022.
(**Outstanding Paper Award**)

L. Weihs, A. Yuile, R. Baillargeon, C. Fisher, G. Marcus, **R. Mottaghi**, A. Kembhavi. **Benchmarking Progress to Infant-Level Physical Reasoning in AI**, Transactions on Machine Learning Research (TMLR), 2022.

D. Schwenk, A. Khandelwal, C. Clark, K. Marino, **R. Mottaghi**. **A-OKVQA: A Benchmark for Visual Question Answering using World Knowledge**, European Conference on Computer Vision (ECCV), 2022.

P. Chattopadhyay, J. Hoffman, **R. Mottaghi**, A. Kembhavi. **RobustNav : Towards Benchmarking Robustness in Embodied Navigation**, in International Conference on Computer Vision (ICCV), 2021. (**Oral presentation**)

M. Deitke*, W. Han*, A. Herrasti*, A. Kembhavi*, E. Kolve*, **R. Mottaghi***, J. Salvador*, D. Schwenk*, E. Vanderbilt*, M. Wallingford*, L. Weihs*, M. Yatskar*, and A. Farhadi. **RoboTHOR: An Open Simulation-to-Real Embodied AI Platform**, in Conference on Computer Vision and Pattern Recognition (CVPR), 2020.
(* Alphabetically listed equal contribution)

M. Shridhar, J. Thomason, D. Gordon, Y. Bisk, W. Han, **R. Mottaghi**, L. Zettlemoyer, and D. Fox. **ALFRED: A Benchmark for Interpreting Grounded Instructions for Everyday Tasks**, in Conference on Computer Vision and Pattern Recognition (CVPR), 2020.

K. Marino, M. Rastegari, A. Farhadi, and **R. Mottaghi**. **OK-VQA: A Visual Question Answering Benchmark Requiring External Knowledge**, in Conference on Computer Vision and Pattern Recognition (CVPR), 2019.

P. Anderson, A. Chang, D. Chaplot, A. Dosovitskiy, S. Gupta, V. Koltun, J. Kosecka, J. Malik, **R. Mottaghi**, M. Savva, and A. R. Zamir. **On Evaluation of Embodied Navigation Agents**, arXiv, 2018.

Y. Xiang, W. Kim, W. Chen, J. Ji, C. Choy, H. Su, **R. Mottaghi**, L. Guibas, and S. Savarese. **ObjectNet3D: A Large Scale Database for 3D Object Recognition**, in European Conference on Computer Vision (ECCV), 2016.

R. Mottaghi, X. Chen, X. Liu, S. Fidler, R. Urtasun, A. Yuille. **The Role of Context for Object Detection and Semantic Segmentation in the Wild**, in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2014.

X. Chen, **R. Mottaghi**, X. Liu, N. Cho, S. Lee, S. Fidler, R. Urtasun, A. Yuille. **Detect What You Can: Detecting and Representing Objects using Holistic Models and Body Parts**, in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2014.

Y. Xiang, **R. Mottaghi**, S. Savarese. **Beyond PASCAL: A Benchmark for 3D Object Detection in the Wild**, in IEEE Winter Conference on Applications of Computer Vision (WACV), 2014.